Overview of project task

TANA21/22: Beräkningsmatematik | *Andrew R. Winters* | *HT1 2026*

There is a project component to this introductory course in scientific computing. The learning goals of this project are to develop the abilities to:

1. Translate a mathematical formula or algorithm into a computer program.
2. Debug a computer program.
3. Verify that the implementation of a formula or algorithm is correct by comparing computed results with theoretical estimates.
4. Write a scientific report that collects and discusses your results from the computer project.

Overall, the project task serves to unify scientific computing concepts from the lecture, the written exercises, and the computer exercises.

Organization and presenting results

The project component is divided into milestones such that tasks pertinent to the current Lecture subject(s) can be worked on in tandem. The number of milestones a student must complete depends on their course registration:

- Students registered in TANA21 will have two milestones.
- Students registered in TANA22 and 9AMA74 will have four milestones.

The project to be solved and the final project report will be written in same groups in which students complete the computer lab exercises.

For students registered in TANA21 the project comes in two "flavours" called `Version Eins` or `Version Zwei`. Each pair of students solve one of the versions that will be assigned through the course homepage. This assignment also determines how pairs of students will oppose one another during presentation in the seminar portion. Students registered in TANA22 or 9AMA74 will complete the project "flavours" `Version Drei` or `Version Vier` and its milestones. A similar opposition pair strategy for the seminar will be taken for this project version as well.

There will be a Handledning session on at the beginning of Week 41 for assistance with the project and its milestones. The Seminar is then held at the end of Week 42 on Thursday or Friday (depending on your Group). Note, at this seminar you will present only a draft of your final report. The complete report is not due until one week after the seminar session.

To describe the numerical problem and results obtained, the pairs of students must prepare a written project report. This project report can be subdivided to discuss each mini-project separately if it helps with organization. Details on the style and content for the project report is provided on the course homepage in the document `project_writeup_format.pdf`.

Your project report can be written in either English or Swedish.

Preparation and submission of project task

Each of the project (whichever version your group has been assigned) are solved in following way:

1. Read the project description with particular focus on any formula, algorithms, or theoretical results that are of importance.
2. Solve the problem in python.
3. Perform the numerical tasks that answer the questions posed by the particular project.
4. Compile necessary numerical results and write the report in the required format. Your project report can be written in either English or Swedish.
5. Submit the written report to the opposing group at least one day before the seminar session.
6. Prepare for the opposition (described below).
7. Print a paper copy of your own report and bring it to the seminar, although having your computer available is also useful.
8. Discuss the project during the seminar in your assigned opposition groups forming discussion pairs.
9. Make any changes to your report based on this discussion and submit them to the seminar leader within a week of the seminar date.

Use the checklists below to prepare your own report as well as the opposition to another group's report.

Verify that:

☐ The report is written according to the required format.

☐ The numerical results and their discussion answer the questions posed in the project description.

Consider if:

☐ You know that the report clearly explains what you did, how you did it, and if you resolved the stated problems.

☐ You know that the report convinces yourself and others that the solutions and your python codes are correct.

☐ You think it would be straightforward to replicate the solution procedure and obtain similar results to those presented in the report.